

A General-Coefficient Conjugate Potential

Full affirmative answer to a question in arXiv:1012.1735

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July 22, 2026

Status: candidate full solution. The proof is complete and uses only the first-order identities already established in the source paper. Independent expert review is recommended, with special attention to the distributional slice argument in Step 3.

1 Source question

The source is Pascal Auscher and Andreas Rosén, *Weighted maximal regularity estimates and solvability of non-smooth elliptic systems II*, arXiv:1012.1735, Section 3, immediately after Proposition 3.5, PDF page 16 [1].

These equations hold in $\mathbf{O}^{1+n} \setminus \{0\}$. Next, applying D to (19) yields

$$(\partial_t + DB + \frac{n-1}{2}N)(Dv) = 0.$$

Thus $f := Dv$ satisfies (17) and $f_t \in R(D) = \mathcal{H}$. By Corollary 3.4 there is a weak solution $\tilde{u}$ in $\mathbf{O}^{1+n}$ of the divergence form equation associated to f . In particular, $f_{\parallel} = r^{(n+1)/2}(r^{-1}\nabla_S \tilde{u})$ and $f_{\perp} = r^{(n+1)/2}(A\nabla_{\mathbf{x}} \tilde{u})_{\perp}$. Applying the conormal gradient-to-gradient map, we deduce $\nabla_{\mathbf{x}} \tilde{u} = \nabla_{\mathbf{x}} u$ in $\mathbf{O}^{1+n} \setminus \{0\}$ distribution sense. In particular, $u = \tilde{u} + c$ in $\mathbf{O}^{1+n} \setminus \{0\}$ for some constant c . As $\tilde{u} + c$ is also a weak solution in $\mathbf{O}^{1+n}$ to the divergence form equation with coefficients A , this provides us with the desired extension for u . $\square$

For perturbations A of radially independent coefficients, Corollary 12.8(i) proves a converse of this result, *i.e.* the existence of such a vector-valued potential v containing a given solution u to $\operatorname{div}_{\mathbf{x}} A \nabla_{\mathbf{x}} u = 0$ as normal component. We do not know whether such v can be defined for general coefficients (except in $\mathbf{O}^2$, see Section 5).

Remark 3.6. Assume that the coefficients A are defined in $\mathbf{R}^{1+n}$ and that the accretivity condition (2) or (14) holds for a.e $r \in (0, \infty)$. As in Proposition 3.3 there is also a one-to-one correspondence between solutions $g \in L_2^{loc}(\mathbf{R}^{1+n} \setminus \overline{\mathbf{O}^{1+n}}; L_2(S^n; \mathcal{V}))$ to $\operatorname{div}_{\mathbf{x}}(Ag) = 0$, $\operatorname{curl}_{\mathbf{x}} g = 0$ in the exterior of the unit ball and solutions $f : \mathbf{R}_{-} \rightarrow \mathcal{H}$ to the equation $\partial_t f + (DB + \frac{n-1}{2}N)f = 0$ for $t < 0$ in $L_2(\mathbf{R}_{-}; \mathcal{H})$. Also, as in Proposition 3.5, *loc* solutions $g : \mathbf{R}^{1+n} \setminus \overline{\mathbf{O}^{1+n}} \rightarrow L_2(S^n; \mathbf{O}(D))$ to the equation $\partial_t g + (DB + \frac{n-1}{2}N)g = 0$

Figure 1: The open question in the source paper, PDF page 16.

The question is whether, for general bounded strictly accretive coefficients, every weak solution u admits a vector-valued potential v containing u as its normal component and satisfying the companion first-order system. The source proves this for perturbations of radially independent coefficients and, separately, in the two-dimensional case.

2 Definitions and notation

We retain the notation of [1, Section 3]. Put

$$\sigma = \frac{n-1}{2}, \quad r = e^{-t},$$

and let

$$D = \begin{bmatrix} 0 & -\operatorname{div}_S \\ \nabla_S & 0 \end{bmatrix}, \quad N = \begin{bmatrix} -I & 0 \\ 0 & I \end{bmatrix}.$$

Thus $ND = -DN$ and

$$\mathcal{H} = \operatorname{ran} D = \begin{bmatrix} \operatorname{ran} \operatorname{div}_S \\ \operatorname{ran} \nabla_S \end{bmatrix}.$$

Let A be a coefficient field satisfying the hypotheses of Proposition 3.3 of [1], and write $B = \hat{A}$ for its transformed coefficient field. If u is a weak solution of

$$\operatorname{div}_{\mathbf{x}}(A\nabla_{\mathbf{x}}u) = 0$$

in the unit ball, its conormal gradient f belongs to $L^2_{\operatorname{loc}}(\mathbb{R}_+; \mathcal{H})$ and satisfies

$$\partial_t f + (DB + \sigma N)f = 0 \tag{1}$$

distributionally. Moreover,

$$f_{\parallel} = r^{\sigma} \nabla_S u_r, \quad (Bf)_{\perp} = r^{\sigma+1} \partial_r u_r. \tag{2}$$

A vector potential associated to u is a section $v = (v_{\perp}, v_{\parallel})$ such that

$$\partial_t v + (BD - \sigma N)v = 0, \quad Dv = f, \quad v_{\perp} = r^{\sigma} u_r. \tag{3}$$

The source additionally requires $v \in L^2_{\operatorname{loc}}(\mathbb{R}_+; \operatorname{dom} D)$ and $\int_1^{\infty} \|Dv_t\|_2^2 dt < \infty$.

3 Proof intuition

The normal component of v is forced: set $v_{\perp} = r^{\sigma} u_r$. The tangential component has two apparent duties. Its divergence must equal $-f_{\perp}$, and it must solve the tangential component of the v -equation. These duties are compatible because the normal component of the already-known f -equation is exactly the evolution equation for the divergence constraint. Hence one chooses the tangential field at a single time slice and evolves it by a scalar-coefficient Bochner ODE. No perturbation theory, boundary trace, or inversion of the singular first-order operator is needed.

4 Full result

Theorem 1 (General-coefficient converse to Proposition 3.5). *Let A and $B = \hat{A}$ satisfy the hypotheses of Proposition 3.3 of [1], and let u be any weak solution of $\operatorname{div}_{\mathbf{x}}(A\nabla_{\mathbf{x}}u) = 0$ in the unit ball. Then there exists*

$$v \in L^2_{\operatorname{loc}}(\mathbb{R}_+; \operatorname{dom} D)$$

such that

$$\partial_t v + (BD - \sigma N)v = 0, \quad Dv = f, \quad u_r = r^{-\sigma} (v_t)_{\perp},$$

where f is the conormal gradient of u . Furthermore,

$$\int_1^\infty \|Dv_t\|_2^2 dt < \infty.$$

Thus the vector-valued potential asked for in [1, p. 16] exists for arbitrary coefficients covered by Proposition 3.3.

Proof. Write $f = (f_\perp, f_\parallel)$ and $Bf = ((Bf)_\perp, (Bf)_\parallel)$.

Step 1: fix the normal component. Define

$$U_t(x) := r^\sigma u_r(x), \quad r = e^{-t}.$$

The component identities (2) give, distributionally on $\mathbb{R}_+ \times S^n$,

$$\nabla_S U_t = f_{t,\parallel} \tag{4}$$

and

$$\partial_t U_t + (B_t f_t)_\perp + \sigma U_t = 0. \tag{5}$$

Indeed, $\partial_t = -r\partial_r$, so

$$\partial_t(r^\sigma u_r) = -\sigma r^\sigma u_r - r^{\sigma+1} \partial_r u_r.$$

Step 2: choose one tangential slice. Since $f_t \in \mathcal{H}$ for almost every t , its normal component belongs to ran div_S . Choose a time $t_* > 0$ at which $f_{t_*,\perp} \in L^2(S^n; \mathbb{C}^m)$ is defined. The range of div_S is closed on the sphere, so there exists

$$w_* \in \text{dom}(\text{div}_S) \quad \text{such that} \quad -\text{div}_S w_* = f_{t_*,\perp}.$$

Step 3: evolve the tangential component. For every $t > 0$, define the Bochner integral

$$w_t = e^{\sigma(t-t_*)} w_* - \int_{t_*}^t e^{\sigma(t-s)} (B_s f_s)_\parallel ds, \tag{6}$$

with the usual signed interpretation when $t < t_*$. Since $Bf \in L_{\text{loc}}^2(\mathbb{R}_+; L^2)$, this defines $w \in W_{\text{loc}}^{1,2}(\mathbb{R}_+; L^2)$ and

$$\partial_t w_t + (B_t f_t)_\parallel - \sigma w_t = 0. \tag{7}$$

It remains to check that the divergence constraint is preserved. The normal component of (1) is

$$\partial_t f_{t,\perp} - \text{div}_S(B_t f_t)_\parallel - \sigma f_{t,\perp} = 0. \tag{8}$$

This identity is valid in $W^{-1,2}(S^n)$ and implies the variation-of-constants formula

$$f_{t,\perp} = e^{\sigma(t-t_*)} f_{t_*,\perp} + \int_{t_*}^t e^{\sigma(t-s)} \text{div}_S(B_s f_s)_\parallel ds. \tag{9}$$

Applying $-\text{div}_S$ to (6) and using $-\text{div}_S w_* = f_{t_*,\perp}$ gives precisely the right-hand side of (9). Therefore

$$-\text{div}_S w_t = f_{t,\perp} \tag{10}$$

in $W^{-1,2}$ for every admissible representative, and in L^2 for almost every t . In particular $w_t \in \text{dom div}_S$ for almost every t .

Step 4: assemble the potential. Set

$$v_t := \begin{bmatrix} U_t \\ w_t \end{bmatrix}.$$

Equations (4) and (10) yield

$$Dv_t = \begin{bmatrix} -\operatorname{div}_S w_t \\ \nabla_S U_t \end{bmatrix} = \begin{bmatrix} f_{t,\perp} \\ f_{t,\parallel} \end{bmatrix} = f_t.$$

The normal component of $\partial_t v + (BD - \sigma N)v$ vanishes by (5); its tangential component vanishes by (7). Thus (3) holds distributionally. By construction,

$$u_r = r^{-\sigma}(v_t)_\perp.$$

Finally, U , w , and $Dv = f$ are locally square integrable in the graph norm of D . Corollary 3.4 of [1] gives

$$\int_1^\infty \|Dv_t\|_2^2 dt = \int_1^\infty \|f_t\|_2^2 dt < \infty.$$

All requirements of Proposition 3.5 and the source question are satisfied. $\square$

5 Verification notes

1. **Sign check.** Since $N(U, w) = (-U, w)$, the two components of the desired equation are $U' + (Bf)_\perp + \sigma U = 0$ and $w' + (Bf)_\parallel - \sigma w = 0$, exactly (5) and (7).
2. **Compatibility check.** Applying $-\operatorname{div}_S$ to the tangential evolution produces the normal equation (8); this is the central identity and uses no pointwise ellipticity.
3. **Domain check.** Equality (10) upgrades w_t from L^2 to $\operatorname{dom} \operatorname{div}_S$ for almost every t , while (4) gives $U_t \in \operatorname{dom} \nabla_S$.
4. **Endpoint check.** The construction is local in t and therefore needs no trace at $t = 0$. The only global condition is on $Dv = f$ for $t > 1$, already supplied by the source paper.
5. **Nonuniqueness.** A different choice of w_* changes w_t by $e^{\sigma(t-t_*)}h$ with $\operatorname{div}_S h = 0$. Existence, not uniqueness, is the question asked.

6 Novelty search, limitations, and review recommendation

A bounded search on July 22, 2026 used the exact sentence from the question, the arXiv identifier, and combinations of “vector-valued potential”, “general coefficients”, “conormal gradient”, and the two first-order systems. It found the source paper and its published version, but no later source claiming this general converse. The proof above is elementary relative to the source framework, so the main novelty risk is that the observation may exist under different terminology.

The theorem addresses only the existence question immediately following Proposition 3.5. It does not claim the boundary estimates, trace regularity, or perturbative norm bounds later obtained in the source under Carleson hypotheses. The recommended human review should focus on the $W^{-1,2}$ variation-of-constants step and on whether the authors intended any unstated global bound on v beyond the explicit hypotheses of Proposition 3.5.

References

- [1] P. Auscher and A. Rosén, *Weighted maximal regularity estimates and solvability of non-smooth elliptic systems II*, Analysis & PDE **5** (2012), no. 5, 983–1061; arXiv:1012.1735.